

Lists, Multimedia, and Tables: Organizing Web Content

Introduction: Structuring Information on the Web

A web page without structure is like a library without shelves — all the information exists, but it is difficult to find, read, or understand. HTML provides powerful tools for organizing content in ways that are both visually clear and semantically meaningful.

Three of the most important tools for structuring web content are **lists**, **multimedia elements**, and **tables**. Lists help organize information into ordered or unordered groups. Multimedia elements allow you to embed images, videos, and audio to make pages more engaging. Tables provide a structured way to display data in rows and columns.

This reading material examines how to use HTML lists (ordered, unordered, and definition lists), embed multimedia elements (images, video, and audio), and create tables with various attributes and values. By the end of this lesson, learners will be able to create well-structured web pages that present information clearly and effectively.

Part 1: HTML Lists – Organizing Information

Introduction to Lists

HTML lists allow you to group related items together. They are essential for creating navigation menus, displaying steps in a process, listing features, and organizing content in a readable format.

There are **three types of lists** in HTML:

1. **Unordered Lists** – Bulleted lists (no specific order)
2. **Ordered Lists** – Numbered lists (items have a specific sequence)
3. **Definition Lists** – Lists of terms and their definitions

Unordered Lists ()

An **unordered list** displays items with bullet points. The order of items does not matter.

Syntax:

```
html
<ul>
  <li>First item</li>
  <li>Second item</li>
  <li>Third item</li>
</ul>
```

Output:

- First item
- Second item
- Third item

List Styles (Attributes):

The `type` attribute changes the bullet style:

Attribute Value

Bullet Style

`disc`

Solid circle (default)

`circle`

Hollow circle

`square`

Solid square

```
html
<ul type="square">
  <li>Item with square bullet</li>
</ul>
```

Ordered Lists ()

An **ordered list** displays items with numbers or letters. The order of items is important.

Syntax:

```
html
<ol>
  <li>First step</li>
  <li>Second step</li>
  <li>Third step</li>
</ol>
```

Output:

1. First step
2. Second step
3. Third step

List Styles (Attributes):

The `type` attribute changes the numbering style:

Attribute Value	Numbering Style
<code>1</code>	Numbers (1, 2, 3) – default
<code>A</code>	Uppercase letters (A, B, C)
<code>a</code>	Lowercase letters (a, b, c)
<code>I</code>	Uppercase Roman numerals (I, II, III)
<code>i</code>	Lowercase Roman numerals (i, ii, iii)

```
html
<ol type="A">
  <li>Item A</li>
  <li>Item B</li>
  <li>Item C</li>
</ol>
```

Output:

- A. Item A
- B. Item B
- C. Item C

The `start` attribute specifies the starting number:

```
html
<ol start="5">
  <li>Starts at 5</li>
  <li>Continues to 6</li>
</ol>
```

Nested Lists

Lists can be nested inside each other to create sub-lists.

Example:

```
html
<ul>
  <li>Fruits
    <ul>
      <li>Apple</li>
      <li>Banana</li>
      <li>Mango</li>
    </ul>
  </li>
  <li>Vegetables
    <ul>
      <li>Carrot</li>
      <li>Broccoli</li>
    </ul>
  </li>
</ul>
```

Output:

- Fruits
 - Apple

- Banana
- Mango
- Vegetables
- Carrot
- Broccoli

Definition Lists (<dl>)

A **definition list** is used for terms and their definitions. It consists of:

- <dl> – The definition list container
- <dt> – The definition term
- <dd> – The definition description

Syntax:

```
html
<dl>
  <dt>HTML</dt>
  <dd>HyperText Markup Language – the standard language for creating web pages</dd>
  <dt>CSS</dt>
  <dd>Cascading Style Sheets – used to style HTML elements</dd>
</dl>
```

Output:

HTML
: HyperText Markup Language – the standard language for creating web pages

CSS
: Cascading Style Sheets – used to style HTML elements

Part 2: HTML Multimedia – Enhancing Web Pages

Introduction to Multimedia

Multimedia elements make web pages more engaging and interactive. HTML5 introduced native support for embedding **images**, **video**, and **audio** without requiring third-party plugins.

Images ()

The `<img>` tag is used to embed images in a web page. It is an **empty element** (no closing tag).

Syntax:

```
html

```

Key Attributes:

Attribute	Description
<code>src</code>	Specifies the image file path or URL
<code>alt</code>	Alternative text displayed if the image fails to load (required for accessibility)
<code>width</code>	Specifies the width of the image (in pixels or percentage)
<code>height</code>	Specifies the height of the image (in pixels or percentage)

Example:

```
html

```

Video (<video>)

The <video> tag is used to embed video files. Multiple source formats can be provided for browser compatibility.

Syntax:

```
html
<video width="640" height="360" controls>
  <source src="video.mp4" type="video/mp4">
  <source src="video.webm" type="video/webm">
  Your browser does not support the video tag.
</video>
```

Key Attributes:

Attribute	Description
controls	Displays play/pause and volume controls
autoplay	Starts playing automatically when the page loads
loop	Repeats the video continuously
muted	Mutes the audio
poster	Displays an image while the video loads

Audio (<audio>)

The <audio> tag is used to embed audio files.

Syntax:

```
html
<audio controls>
  <source src="audio.mp3" type="audio/mpeg">
  <source src="audio.ogg" type="audio/ogg">
  Your browser does not support the audio tag.
```

</audio>

Key Attributes:

Attribute	Description
controls	Displays play/pause and volume controls
autoplay	Starts playing automatically
loop	Repeats the audio continuously
muted	Mutes the audio

Common Multimedia File Formats

Media Type	Supported Formats
Images	JPEG, PNG, GIF, SVG, WebP
Video	MP4, WebM, OGG
Audio	MP3, WAV, OGG, AAC

Part 3: HTML Tables – Displaying Data

Introduction to Tables

HTML tables are used to display data in rows and columns. They are essential for presenting structured information such as schedules, price lists, comparison charts, and statistical data.

Basic Table Structure

A table consists of:

- `<table>` – The table container
- `<tr>` – Table row
- `<th>` – Table header cell (bold and centered by default)
- `<td>` – Table data cell (regular cell)

Syntax:

```
html
<table>
  <tr>
    <th>Name</th>
    <th>Age</th>
    <th>City</th>
  </tr>
  <tr>
    <td>Juan</td>
    <td>17</td>
    <td>Manila</td>
  </tr>
  <tr>
    <td>Maria</td>
    <td>18</td>
    <td>Cebu</td>
  </tr>
</table>
```

Output:

Name	Age	City
Juan	17	Manila
Maria	18	Cebu

Table Attributes

Attribute	Description
<code>border</code>	Specifies the border width (e.g., <code>border="1"</code>)
<code>cellpadding</code>	Space between the cell content and its border
<code>cellspacing</code>	Space between cells
<code>width</code>	Specifies the table width (in pixels or percentage)
<code>align</code>	Aligns the table (left, center, right)

Example with Attributes:

```
html
<table border="1" cellpadding="10" cellspacing="5" width="80%" align="center">
  <tr>
    <th>Product</th>
    <th>Price</th>
  </tr>
  <tr>
    <td>Laptop</td>
    <td>₹45,000</td>
  </tr>
</table>
```

Table Headers and Captions

Table Caption (`<caption>`): Adds a title to the table.

Table Headers (`<th>`): Can be applied to rows or columns to indicate what the data represents.

Example:

html

```
<table border="1">
  <caption>Student Enrollment by Grade Level</caption>
  <tr>
    <th>Grade Level</th>
    <th>Number of Students</th>
  </tr>
  <tr>
    <th>Grade 11</th>
    <td>350</td>
  </tr>
  <tr>
    <th>Grade 12</th>
    <td>320</td>
  </tr>
</table>
```

Output:

Grade Level	Number of Students
Grade 11	350
Grade 12	320

Spanning Rows and Columns

Attribute	Description
colspan	Merges multiple columns into one cell
rowspan	Merges multiple rows into one cell

Example (colspan):

```
html
<table border="1">
  <tr>
    <th colspan="2">Student Information</th>
  </tr>
  <tr>
    <td>Name</td>
    <td>Maria</td>
  </tr>
  <tr>
    <td>Age</td>
    <td>17</td>
  </tr>
</table>
```

Output:

Student Information

Name	Maria
Age	17

Example (rowspan):

```
html
<table border="1">
  <tr>
    <th rowspan="2">Department</th>
    <th>Employee</th>
  </tr>
  <tr>
    <td>Juan Dela Cruz</td>
  </tr>
</table>
```

Output:

Department

Employee

IT

Juan Dela Cruz

Complete Table Example

html

```
<!DOCTYPE html>
```

```
<html>
```

```
  <head>
```

```
    <title>Sample Table</title>
```

```
  </head>
```

```
  <body>
```

```
    <h1>Class Schedule</h1>
```

```
    <table border="1" cellpadding="10">
```

```
      <caption>Grade 11 Schedule - First Semester</caption>
```

```
      <tr>
```

```
        <th>Time</th>
```

```
        <th>Monday</th>
```

```
        <th>Tuesday</th>
```

```
        <th>Wednesday</th>
```

```
        <th>Thursday</th>
```

```
        <th>Friday</th>
```

```
      </tr>
```

```
      <tr>
```

```
        <td>8:00 - 9:00</td>
```

```
        <td>Math</td>
```

```
        <td>English</td>
```

```
        <td>Math</td>
```

```
        <td>Science</td>
```

```
        <td>English</td>
```

```
      </tr>
```

```
      <tr>
```

```
        <td>9:00 - 10:00</td>
```

```
        <td>Science</td>
```

```
        <td>Math</td>
```

```
        <td>English</td>
        <td>Math</td>
        <td>Science</td>
    </tr>
</table>
</body>
</html>
```

Key Terms

Term	Definition
Unordered List	A bulleted list using <code></code> ; order does not matter
Ordered List	A numbered list using <code></code> ; order is important
Definition List	A list of terms and their definitions using <code><dl></code> , <code><dt></code> , and <code><dd></code>
Nested List	A list placed inside another list to create sub-levels
Multimedia	Content that combines text, images, video, and audio
<code></code>	The HTML tag used to embed images
<code><video></code>	The HTML tag used to embed video files
<code><audio></code>	The HTML tag used to embed audio files
Table	An HTML structure that displays data in rows and columns
<code><table></code>	The HTML tag that defines a table
<code><tr></code>	Table row element
<code><th></code>	Table header cell element
<code><td></code>	Table data cell element
<code>colspan</code>	An attribute that merges multiple columns
<code>rowspan</code>	An attribute that merges multiple rows

Review Questions

Write your answers on a separate sheet of paper.

1. What is the difference between an unordered list and an ordered list? Give an example of when you would use each.
2. Write the HTML code to create an ordered list of the three best-selling books using lowercase Roman numerals.
3. Write the HTML code to create a definition list for the terms "HTML," "CSS," and "JavaScript."
4. What is the purpose of the `alt` attribute in an `<img>` tag?
5. Write the HTML code to embed a video file named `intro.mp4` with controls.
6. What is the difference between `colspan` and `rowspan` in an HTML table?
7. Write the HTML code to create a table with three columns (Name, Age, City) and two rows of data.
8. What are the three types of lists available in HTML? Describe each briefly.
9. Write the HTML code for a nested list showing:
 - Fruits (with Apple and Banana as sub-items)
 - Vegetables (with Carrot and Broccoli as sub-items)
10. What attributes are commonly used with the `<table>` tag to control its appearance? Give at least three.

Answer Key

1. An unordered list (`<ul>`) uses bullet points and is used when the order of items does not matter (e.g., shopping list). An ordered list (`<ol>`) uses numbers and is used when the order is important (e.g., step-by-step instructions).

2.

text

```
<ol type="i">
  <li>The Great Gatsby</li>
  <li>To Kill a Mockingbird</li>
  <li>1984</li>
```


3.

text

```
<dl>
  <dt>HTML</dt>
  <dd>HyperText Markup Language</dd>
  <dt>CSS</dt>
  <dd>Cascading Style Sheets</dd>
  <dt>JavaScript</dt>
  <dd>A programming language for web interactivity</dd>
</dl>
```

4. The `alt` attribute provides alternative text that is displayed if the image fails to load. It is also essential for accessibility, allowing screen readers to describe the image to visually impaired users.

5.

text

```
<video controls>
  <source src="intro.mp4" type="video/mp4">
  Your browser does not support the video tag.
</video>
```

6. `colspan` merges multiple columns into one cell (horizontal spanning). `rowspan` merges multiple rows into one cell (vertical spanning).

7.

text

```
<table border="1">
  <tr>
    <th>Name</th>
    <th>Age</th>
    <th>City</th>
  </tr>
  <tr>
    <td>Juan</td>
    <td>17</td>
```

```

        <td>Manila</td>
    </tr>
    <tr>
        <td>Maria</td>
        <td>18</td>
        <td>Cebu</td>
    </tr>
</table>

```

8. Three types of HTML lists:

- Unordered List () – Bulleted list for items in no specific order
- Ordered List () – Numbered list for items in a specific sequence
- Definition List (<dl>) – List of terms (<dt>) and their descriptions (<dd>)

9.

```

text
<ul>
    <li>Fruits
        <ul>
            <li>Apple</li>
            <li>Banana</li>
        </ul>
    </li>
    <li>Vegetables
        <ul>
            <li>Carrot</li>
            <li>Broccoli</li>
        </ul>
    </li>
</ul>

```

10. Common table attributes:

- border – Specifies border width
- cellpadding – Space between cell content and border
- cellspacing – Space between cells
- width – Table width (pixels or percentage)
- align – Table alignment (left, center, right)